Intro To High-performance Computing

Workshop Lead: Megan Ng

Registration link: https://forms.cloud.microsoft/r/xJjdKkVTUC

Approximate duration: 2 hours

Prerequisites:

1. A Digital Research Alliance of Canada account and access to one of their general-purpose clusters (Fir, Narval, Nibi, or Rorqual)
2. A terminal (Mac OS and any linux distribution will have a terminal already but Windows users must download their own terminal)
3. The Duo Mobile application downloaded on your phone
4. Globus Connect Personal software downloaded on your laptop
5. A code editor (The workshop lead will be using VS Code)

Summary:

This workshop is designed to give you practical experience using the high-performance computing system offered by the Digital Research Alliance of Canada (DRAC). Through guided hands-on activities, you can expect to leave this workshop with the knowledge and skills required run your own computational experiments using DRAC start to finish – from organizing and uploading your data and code to submitting jobs, and troubleshooting common errors.

Learning Objectives:

1. Conceptually understand the workflow of using DRAC
2. Learn how to use DRAC start to finish
   a. Uploading data and code to DRAC storage systems
   b. Running code using DRAC high-performance computers
   c. Retrieving and assessing results
3. Identify and handle common errors that occur when using DRAC

Content:

1. **Module I: About the Digital Research Alliance of Canada (DRAC) (15min)**
   a. What is the DRAC and the services they offer
   b. Why do researchers use DRAC services

c. Workflow description of how you interact with DRAC's high-performance computing system
d. Hands-on activity: DRAC Workflow understanding quiz (5min)

2. **Module II: Preparing Your Data and Code for DRAC (30min)**
   a. Best practices for preparing your data and code to be run on DRAC
   b. Introduction to DRAC's file transfer system (Globus)
   c. Hands-on activity: Upload workshop data and code to DRAC using Globus Personal Connect (25min)
      i. Uploading files to DRAC
      ii. Navigating folders with linux

3. **Module III: Running a Job on DRAC (30min)**
   a. DRAC's file system manager (slurm)
   b. How to communicate your computational needs to DRAC
      i. What is a job script
      ii. What information must you provide in a job script
   c. How to "submit a job"
   d. How to check on the status of your job
   e. Hands-on activity: Submit a job to DRAC (20min)

4. **Module IV: Handling Job Errors (30min)**
   a. How to use output and error files
   b. Common errors and their solutions
   c. Practical DRAC debugging advice
   d. DRAC technical support resources
   e. Hands-on activity: Identify and address two most common errors (20min)